

WARM-UP AND SKILL WORK

CLASSIFICATION OF WARM-UP EXERCISES AND DESCRIPTIONS

The goal of a warm-up is to prime the body into an optimal state for your workout and address any deficiencies in movement. Constructing a universal warm-up is difficult since every athlete's body has different needs. The three essential components of a warm-up involve blood flow, mobility, and drilling good body positioning—how long you will spend on each component will vary according to your own body's needs.

Sample Warm-Up Routine

(Will take approximately 15-20 minutes to complete.)

Blood Flow

- 10-20 Burpees
- 60s of Crawling

Mobility

- 10x Wrist Circles
- 10x Shoulder Circles
- 10x Bodyweight Squats
- Any other body part or joint that needs to be warmed up
- 60s of Support Hold Work (using parallel bars/rings or chairs/counters)
- 5x German Hangs / Skin the Cat

Positional (Bodyline/Body Tension) Drills

- 30-60s of Plank, both Side Planks, Reverse Plank, Hollow, Arch Holds

Skill Work: Time of skill work, type of skill work, and qualities

- 5-10m of Handstand Work (against the wall)

Blood Flow

A few key physiological changes must happen in the body to ensure a good warm-up. You must raise core temperature so the chemical reactions in your muscles take place faster, leading to better contractile function

and activation of the nervous system. Additionally, your heart rate and blood flow to the muscles should be elevated, in order to provide oxygen and nutrients and export waste. Other physiological changes that must take place include: increasing blood perfusion to the muscles, tendons, and other connective tissue, priming the brain and nervous system for exercise, and ensuring that the cartilage in the joints saturates with synovial fluid.

Here is an example of a set of exercises that would constitute a good *blood-flow* warm-up portion of a typical workout:

- 10-20 Burpees
- 60s of Crawling (or crawling for a specific distance, such as 100 meters)

Burpees are an excellent choice because they are a full-body exercise that rapidly increases your heart rate. Additionally, they are easy to perform and carry a low risk of injury.

Burpees are performed as follows in quick sequence: plank position → pushup → plank → move to squat position → stand → jump → squat down → move to plank position and then repeat the entire sequence. Do not pause between motions. The entire exercise should move smoothly. This is a full-body exercise that allows you to practice basic body movements such as lifting yourself up, squatting, and jumping. The fact that burpees are also good for warming up and getting the blood flowing is an added bonus.

The recommendation of sixty seconds of crawling can be broken up into as many sets of sixty seconds as you need. Crawling is performed in the quadruped position with your hands and knees on the ground. You then lift your knees a couple inches off the ground and focus on crawling forward, alternating with opposing limbs. Your left arm and right leg move at the same time and your right arm and left leg move at the same time. Attention should also be given to keeping your back parallel to the ground. Crawling is excellent for activating the core musculature and the scapular stabilizers as well as warming up your entire body. It works a wide range of muscles. To improve rapidly, work on specific distances rather than time—25 meters is enough to make most beginners lightly perspire.

Other types of full-body warm-up exercises to get your blood flowing can include: rolling, squatting, bear walks, crab walks, jumping jacks, jump rope, short jogs, or any other type of movement that is low to moderate intensity, easy to perform, and carries a low risk of injury.

The three main physiological signs that indicate when a person is ready to progress from the blood flow portion of the warm-up to the mobility portion are: 1) a light sweat, 2) moderate increases in heart rate, and 3) a slight to moderate increase in breathing cadence. It is an added bonus when the warm-up exercises lead into mobility work, which is the case with burpees and crawling.

Mobility

The second critical component of a warm-up is *mobility work*. A quick, short circuit of movements that focus on full range of movement to warm up the joints and surrounding tissues is the most useful. In Chapter 4, we covered the difference between flexibility, passive mobility work, and active mobility/flexibility work. For your convenience, the definitions are included again here:

- **Flexibility** – Increasing the range of motion by stretching the muscles through relaxation of the nervous system. The most easily understood example of flexibility to increase range of motion is working to obtain the splits.

- **Passive Mobility** – Taking a joint through its range of motion, but without the aim to increase the range of motion. Passive mobility is a good way to begin to warm up your joints for a workout. For instance, put your wrists on the floor and move your body over them. This will take your wrists to the edge of the range of motion without contracting the wrist muscles and is a great example of passive mobility.
- **Active Mobility/Flexibility** – These two terms are often used interchangeably, with *active flexibility* being the most common. In reality, *active mobility* should be used, since range of motion must be gained through flexibility training prior to actual use. Once this is accomplished, using the new range would be called *mobility training*. An example of this would be using a pike or a straddle stretch to improve compression; actively engaging abdominals and hip flexors to move your face closer to your knees or the ground. Likewise, working a standing split or a kick with your leg above your head is an example of active mobility or flexibility. We will use the term active flexibility as that is the common term.

Here is an example of a standard set of warm-up exercises designed to increase mobility:

- 10x Wrist Circles
- 10x Shoulder Circles
- 10x Bodyweight Squats
- Any other body part or joint that needs to be warmed up
- 60s of Support Hold Work (using parallel bars/rings or chairs/counters)
- 5x German Hangs / Skin the Cat

Most athletes can skip straight to active mobility work, warming up with exercises such as wrist circles, shoulder circles, bodyweight squats, etc. However, if you have particular limitations or stiffness in the joints it may be better to begin with passive mobility before moving into active mobility work. This is also good for older or injured populations. For instance, if your wrists are sore or tight you can move into a quadruped position—on your hands and knees—then move your wrists through various movements, slowly shifting weight back and forth on them to the edge of the range of motion passively. After a couple of minutes of taking the wrist through passive mobility, you may want to start doing active mobility movements such as wrist circles to further warm them up for exercise.

Try a combination of wrist circles (fifteen times each way) and wrist mobility, all ways, on the floor, both flexed and extended. For the shoulders, use a band or stick to do shoulder dislocates. This will help mobilize the scapulae and all of the muscles around the shoulder joint. If you have other methods of mobility or dynamic flexibility work you prefer, feel free to use them instead. This portion of the warm-up should take sixty to ninety seconds. Another facet of mobility work for bodyweight training is the need to warm up the elbows, particularly in the straight-arm position. When progressing from bodyweight exercises into more advanced strength skills the integrity of the elbows is critical. It is necessary to strengthen the biceps to avoid hyperextension.

One to three sets of straight-arm-locked support work is highly recommended. If you are strong enough to do a one-minute hold, that is enough. Try to progress toward turning your palms forward during the hold—what gymnasts call “rings-turned-out” or *RTO* for short. Turning the rings out and performing longer holds in that position is great for your biceps and elbows, as well as warming up the majority of your shoulder muscles.

- If you cannot hold a straight-arm position at all, start on parallettes or parallel bars.
- If you cannot yet hold a straight-arm support on the rings, use a rubber theraband to help you hold the rings together or get someone to spot assist you until you can hold it. This will be necessary for most beginners.
- If you just started training holding a non-RTO support hold, then jump up with straight-arms and try to accumulate between 60-90 seconds total hold time.
- If you are beyond that stage, try to use as few sets as possible to reach the one-minute mark.
- Once you start trying to turn the rings out, be consistent—do not let them come back in.
- Once you have worked your way up to 60-90 seconds with RTO where the rings are in line with your body then set your rings a bit wider—this will make it substantially harder.

The main thing is that not just the muscles are prepared for exercise but also that the tendons, ligaments, cartilage, and joints are as well. This may require more repetitions and/or sets for some people than others. You should stop when your muscles start shaking significantly or when you begin to feel significant pressure in the biceps. Remember, this is a warm-up—taxing yourself now will hinder your workout! Do, however, bear in mind that adding this skill to your warm-up will help you progress more quickly in your strength training. Proficiency on the rings is built by logging a lot of time on the rings.

Once the ring supports are complete, you will want to dynamically stretch your shoulders with some German hangs otherwise known as “skin the cat.” If you are a beginner, set your rings low, let yourself into the position, and stretch for five to ten seconds. If you are more experienced, you can pull out in a tuck or pike position to an inverted hang and then back again. The goal here is to take your shoulders to the edge of their range of motion in extension. This helps with flexion as well: German hangs stretch the chest and lats as well as the anterior shoulder girdle.

Never stretch statically for more than fifteen seconds. Stretching for longer than this may decrease your strength output during your workout. Three to four short stretches should take you about a minute.

Earlier we discussed the length of a warm-up in light of the athlete’s age. Keep in mind that the number of sets and repetitions in your warm-up is somewhat arbitrary. Generally, those who are older or recovering from an injury will require more time to warm-up. Take your age divided by four to see how many minutes you might need to add to your warm-up routine. Use this chart only as a guideline—modify it depending on your circumstances and your personal preferences.

- 20-year-old: 5-10 minutes + $(20/4) = 5-10 + 5$ minutes = 10-15 minutes
- 40-year-old: 5-10 minutes + $(40/4) = 5-10 + 10$ minutes = 15-20 minutes
- 60-year-old: 5-10 minutes + $(60/4) = 5-10 + 15$ minutes = 20-25 minutes

Remember that your warm-ups should be modified as you get stronger. For example, as dips and pull-ups become less strenuous and more second-nature, you can begin to add them to your warm-up routine if you desire.

To summarize, it is important to spend a few minutes of your warm-up mobilizing your joints. After that, you can begin to work on exercises that are two to three levels below your current level on the progression charts. For example: If you are working the straddle planche as one of your primary exercises, you can warm up with some tuck planche pushups to help prepare your muscles for more intense exercise.

Positional Drills

The third critical component of a warm-up is *positional drills* otherwise known as *bodyline* or *body tension drills*. The most useful positioning drills for bodyweight training are the plank, side planks, reverse plank, arches and hollow holds. However, depending on your sport or discipline, there may be other body positional drills that you may potentially use. Select exercises that will facilitate the skill work you require and help you maintain proper body positions in your strength work. Most of these exercises generally fall under the label of *stability work*:

- **Stability Work** – A type of training that focuses primarily on *neuromuscular re-education*, i.e. making your body re-learn how to move properly. For example, when you are recovering from an ankle sprain, one of the primary methods for regaining body awareness or *proprioception*, in the foot and ankle is to work on single leg balances. This activates the muscles around the ankle, which stabilizes it as you re-train your balance. Another example would be using planks as a core stability exercise to train your body to maintain tension in your core in order to hold itself still against gravity.

Stability or body tension drills are performed to solidify positions you will need to know for many bodyweight exercises, as well as to increase stabilization strength in your core. The more you improve at these drills now, the more proficient you will be with your form and technique in bodyweight exercises. This will allow you to progress faster and have a lower risk of being injured.

Many athletes have their own warm-up routines, tailored to their specific mobility weaknesses. Be aware that all bodyweight training places a high demand on connective tissues. Therefore, if you are importing an existing warm-up routine, make sure to also customize it specifically for bodyweight strength training.

So far we have not addressed the role of *stretching*. This is an important component of a workout, particularly because it increases your range of motion. *Static stretching is most beneficial at the end of a workout*. It is most effective when your body is warmed up and your nervous system is fatigued. However, dynamic stretching is useful in a warm-up, as it takes your muscles through their entire range of motion. Unlike static stretching, you do not hold the position at the edge of your range of motion. When static stretches in a warm-up are held for too long, they can have a negative effect on your ability to exert maximal force during your exercise routine. Prior to working out, stretching should be limited, unless you are struggling with poor flexibility that is inhibiting you from properly executing the skills in your routine.

Warming up does not need to be complicated, but it is important that your choices be well constructed so that your body is prepared to execute your routine. Be sure to select warm-up exercises that increase blood flow, mobility, and facilitate body awareness.

A development since the first edition of *Overcoming Gravity* states that it is more effective to include mobility work in your warm-up instead of putting it at the end with prehabilitation and flexibility. The reason for this change is that mobility exercises prepare your body for high-end skill work and your strength development exercises. This means you can intermix basic warm-up exercises with general mobility. Examples of basic warm-up exercises include support holds, easy versions of pushups, rows, etc. Examples of general mobility include German hangs (for the shoulders), wrist mobility work, and bridges and similar movements (for the back). For specific examples of workout structures, refer to the previous chapter of this book.

SKILL WORK: TIME OF SKILL WORK, TYPE OF SKILL WORK, AND QUALITIES

Skill and Technique Work should always take place after your warm-up. This is the optimal time for your body to learn new skills or new movement patterns.

Example: Five to ten minutes of handstand work against the wall. Rest as much as needed between sets of handstands.

Good skill and technique practice should always be emphasized. If you are practicing a skill poorly, your body will memorize this incorrect pattern. An example would be having loose legs (extraneous movement) while performing a handstand. Once your brain and body have memorized an incorrect pattern, it will be difficult to re-adjust your technique. It is always best to focus on correct form from the beginning. If you become too fatigued to continue, the best course of action is to immediately stop and rest. Never give your technique practice a half-effort, as this is usually where incorrect patterns are memorized and poor movements begin.

The planche, front lever, and back lever are *not* skill work. Although handstands and L-sits are included as skill work and all of these exercises are straight-arm skills, keep in mind that skill work is non-fatiguing movements focused on developing specific attributes, such as balance. Using this terminology, we could say that handstand work is essentially “balancing on your hands” work. The planche, front lever, and back lever are incorporated into a routine in order to develop the strength to effectively perform the movements. The proper place for these exercises is in the strength portion of your routine, not in the skill work portion.

On the other hand, skill work is highly variable. Properly performing handstands and various skills such as elbow levers and even L-sits requires a gymnast to spend significant amounts of time in those movements or positions. These skills tend to focus more on balance, but they also require variable amounts of strength. As you become stronger, they become easier.

A note about L-sits: This exercise has been placed in the endurance, core, and isolation work section because most people use this movement for core work, even if it is paired with handstands. L-sits are incredibly fatiguing for many beginners, which leads to decreased capabilities when placed in the strength work section of a workout routine. However, if you wish to train L-sits as skill work, leave them in this portion of your workout. This would be the best course of action if you want to move toward V-sit and manna, where you are not limited by your core strength.

After you master skills like handstands and progressions like pull-ups, dips, and German hangs and they become less intense on your body, they can be used in lieu of a separate warm-up routine. Incorporate them into your warm-up routine only after these exercises have become low intensity. For example, if you have mastered a freestanding handstand and are beginning to work on freestanding handstand pushups, it may be a good idea to add basic handstand work to your warm-up in order to reinforce proper neural patterns for the movement. You can also utilize dynamic movements, such as *handstand shoulder taps* (wall handstands while lifting a hand to touch an alternating shoulder each time) to help warm up your body while training skills at the same time.

All skills or strength progressions that are two to three levels below your current competency level are appropriate exercises for warm-up or skill work, especially if they have a significant balance component, such as handstands and elbow levers. (Use the progression charts as a point of reference.) For example: add short

tuck front lever holds to your warm-up if you are working the straddle front lever progression in your workout. You might also try warming up with an easier version of a progression to help you maintain the movements patterns in which you have become proficient. To save time, you might replace freestanding handstand push-ups with a few wall handstand pushups.

Unlike in barbell training, where you can learn complex movements such as Olympic lifts snatch and the clean and jerk as a beginner and reach a level of proficiency within a few months, this is not possible in bodyweight training. In bodyweight training, the levels of progression are separated based on one's current competency, which takes previous strength and skill development into consideration. For example, the handstand is a basic skill that has a variety of progressions and different levels for one to work through. These include:

- The basic static hold itself, developed from the wall to free standing
- Kicking up and holding a basic handstand
- Handstand walking
- Developing a proper straight-arm press handstand
- Obtaining a freestanding bent-arm handstand pushup
- Obtaining a one-arm handstand
- Controlling various positions in handstands
- One-arm handstands
- One-arm press handstands

Progressing in pure bodyweight work is extremely difficult if you are not under the care of a coach who knows what they are doing and can point you to the correct progressions and offer tips on what you should work on next. Skill development plays a key role in building proper strength. If possible, include skill development in every workout. As an athlete's individual skill, strength, and work capacities improve, exercises previously classified "strength" work may become skill work. If this seems counterintuitive, consider the handstand. Beginners will often have a difficult time holding themselves inverted for five to ten seconds, even with the assistance of a wall. However, as one increases in strength, handstands become an endurance exercise and can be held for sixty seconds or longer. It is important to re-assess your goals and your exercise selections every five to six weeks. You will want to redefine what constitutes skill work and what constitutes strength work on an ongoing basis as your training moves forward.

Skills will not necessarily correlate to level of strength on the progression charts. It is possible to work above or below your strength level when it comes to skill work. Many strong people come into bodyweight training looking only to obtain some of the impressive static positions that are skill-based like the one-arm handstand. With good technique, even relatively weak athletes can work up to one-arm handstands.

Generally speaking, however, the stronger you are, the greater potential you have for skill work. As you get stronger, you will find that your skill work improves, provided that you continue practicing it in conjunction with your strength routine. The key to skill work is the more you practice—leaving adequate time for recovery—the faster your technique will improve. You must find a proper balance that facilitates optimal improvement. For beginners, just getting upside down in a handstand is a difficult feat. But for those who can already hold a freestanding handstand for sixty seconds or more, ten minutes of total work would be possible. Always keep in mind that too much practice can impair recovery, even with skills as simple as handstands.

It is best to approach skill work as you would a warm-up. It should not leave you excessively sweaty or tired when you move into the strength-training portion of your workout. Your goal is to get the highest quality of work without fatigue. For beginners, this may mean your skill work is as light as twenty seconds inverted. On the other hand, those who are more advanced may spend fifteen to twenty minutes practicing difficult skills like the one-arm handstand. Remember, more is not always better. With skill work, err on the side of caution: start with what feels like “too little” and add more if you need it. Those who start with too much skill work will have a difficult time identifying what is wrong if they are not progressing. Put in quality work, but don’t be afraid to quit if you are feeling fatigued or having an off day. There is no shame in knowing that you cannot do something every day; we all have days where nothing goes as planned. Do as much *quality* work as you can per day (as your schedule and level of fatigue allows) and you will improve quickly.

Practice does not make perfect; *perfect* practice makes perfect.

CHAPTER 8 SUMMARY

WARM-UP AND SKILL WORK

A proper warm-up should cover three areas:

- Blood Flow
- Mobility
- Positional Drills

Good blood flow exercises include burpees and crawling, but can also include jumping jacks, jump rope exercises, short jogs, or any other type of movement that is low-to-moderate intensity, low skill level, and carries a low risk of injury. Rolling, squatting, and other quadruped movements such as bear-walks and crab-walks are good, too.

For mobility, choose exercises that warm up every joint. Spend more time on areas that are stiff or have previously been injured. The main recommended long term mobility exercises are RTO supports and German hangs/skin the cats.

The purpose of positional drills is to get your body acclimated to correct positions and maintaining core tension while performing bodyweight exercises. Work up to 30-60 seconds. You can eventually eliminate positional drills as your skills improve. Good position drills are plank, both side planks, reverse plank, hollow and arch holds.

Skill work should cover two areas:

- Sport-specific Work
- Handstands (and other skills that are limited by balance, not strength)

Avoid exercises that are extremely fatiguing in the skill work portion of your routine, as they may decrease your abilities in the latter part of your workout. Otherwise, feel free to work any type of sport-specific drills and movements that require mostly balance components, such as handstands.